

CSA1701-ARTIFICIAL INTELLIGENCE

LAB EXPERIMENTS

5. Missionaries and Cannibals Problem

Aim:

To develop a Python program that solves the Missionaries and Cannibals Problem using BFS to transport 3 missionaries and 3 cannibals across a river safely.

Algorithm:

1. Define states as (missionaries_left, cannibals_left, boat_on_left).
2. Ensure the state is valid: cannibals must never outnumber missionaries on either bank unless missionaries equal zero.
3. Use BFS to explore transitions representing 1 or 2 people crossing the river.
4. Keep track of visited states to avoid cycles.
5. Stop and print the path when the goal state (0, 0, 0) is reached.

```
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from collections import deque

def is_valid(m, c):
    if m < 0 or c < 0 or m > 3 or c > 3: return False
    if 0 < m < c: return False
    if 0 < (3 - m) < (3 - c): return False
    return True

def solve_mc():
    start = (3, 3, 1)
    goal = (0, 0, 0)
    queue = deque([(3, 3, 1)])
    visited = set([start])

    moves = [(1, 0), (2, 0), (0, 1), (0, 2), (1, 1)]

    while queue:
        path = queue.popleft()
        m, c, b = path[-1]

        if (m, c, b) == goal:
            return path

        for dm, dc in moves:
            if b == 1:
                nm, nc, nb = m - dm, c - dc, 0
            else:
                nm, nc, nb = m + dm, c + dc, 1

            if is_valid(nm, nc) and (nm, nc, nb) not in visited:
                visited.add((nm, nc, nb))
                queue.append(path + [(nm, nc, nb)])

    return None

solution = solve_mc()
if solution:
    print("Solution Found! (M, C, Boat)")
    for step in solution:
        print(f"Left Bank: {step[0]} Missionaries, {step[1]} Cannibals | Boat= {'Left' if step[2]==1 else 'Right'}")
else:
    print("No solution exists.")
```

Output:

```
File Edit Shell Debug Options Window Help
Python 3.11.9 (tags/v3.11.9:de54cf5, Apr  2 2024, 10:12:12) [MSC v.1938 64 bit (
AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/Users/VISHAL M/Downloads/ai expl.py
Solution Found! (M, C, Boat)
Left Bank: 3 Missionaries, 3 Cannibals | Boat= Left
Left Bank: 3 Missionaries, 1 Cannibals | Boat= Right
Left Bank: 3 Missionaries, 2 Cannibals | Boat= Left
Left Bank: 3 Missionaries, 0 Cannibals | Boat= Right
Left Bank: 3 Missionaries, 1 Cannibals | Boat= Left
Left Bank: 1 Missionaries, 1 Cannibals | Boat= Right
Left Bank: 2 Missionaries, 2 Cannibals | Boat= Left
Left Bank: 0 Missionaries, 2 Cannibals | Boat= Right
Left Bank: 0 Missionaries, 3 Cannibals | Boat= Left
Left Bank: 0 Missionaries, 1 Cannibals | Boat= Right
Left Bank: 1 Missionaries, 1 Cannibals | Boat= Left
Left Bank: 0 Missionaries, 0 Cannibals | Boat= Right
>>>
```

Result: Thus, the Missionaries and Cannibals Problem solved successfully using Python.